

ascospores develop in and are eventually discharged from the ascocarp.

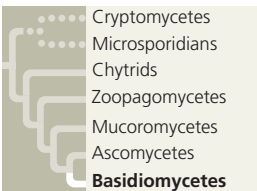
Compared to the life cycle of mucoromycetes, the extended dikaryotic stage of ascomycetes (and also basidiomycetes) provides additional opportunities for genetic recombination. In *Neurospora*, for example, many dikaryotic cells can develop into asci. The haploid nuclei in these asci fuse, and their genomes then recombine during meiosis, resulting in a multitude of genetically different offspring from one mating event (see steps 3–5 in Figure 31.18).

As described in Figure 17.2, biologists in the 1930s used *Neurospora* in research that led to the one gene–one enzyme hypothesis. Today, this ascomycete continues to serve as a model research organism. In 2003, its entire genome was published. This tiny fungus has about three-fourths as many genes as the fruit fly *Drosophila* and about half as many as a human (Table 31.1). The *Neurospora* genome is relatively compact, having few of the stretches of non-coding DNA that occupy so much space in the genomes of humans and many other eukaryotes. In fact, there is evidence that *Neurospora* has a genomic defense system that prevents noncoding DNA such as transposons from accumulating.

Table 31.1 Comparison of Gene Density in *Neurospora*, *Drosophila*, and *Homo sapiens*

	Genome Size (million base pairs)	Number of Genes	Gene Density (genes per million base pairs)
<i>Neurospora crassa</i> (ascomycete fungus)	41	9,700	236
<i>Drosophila melanogaster</i> (fruit fly)	165	14,000	85
<i>Homo sapiens</i> (human)	3,000	<21,000	7

Basidiomycetes



About 50,000 species, including mushrooms, puffballs, and shelf fungi, are called **basidiomycetes** and are classified in the phylum Basidiomycota (Figure 31.19). This phylum also includes mutualists

that form mycorrhizae and two groups of destructive plant parasites: rusts and smuts. The name of the phylum derives from the **basidium** (plural, *basidia*; Latin for “little pedestals”), a cell in which karyogamy occurs, followed immediately by meiosis. The club-like shape of the basidium also gives rise to the common name *club fungus*.

▼ Figure 31.19 Basidiomycetes (club fungi).

► Shelf fungi, important decomposers of wood



► Puffballs emitting spores



► Maiden veil fungus (*Dictyophora*), a fungus with an odor like rotting meat

Basidiomycetes are important decomposers of wood and other plant material. Of all the fungi, certain basidiomycetes are the best at decomposing the complex polymer lignin, an abundant component of wood. Many shelf fungi break down the wood of weak or damaged trees and continue to decompose the wood after the tree dies.

The life cycle of a basidiomycete usually includes a long-lived dikaryotic mycelium. As in ascomycetes, this extended dikaryotic stage provides many opportunities for genetic recombination events, in effect multiplying the result of a single mating. Periodically, in response to environmental stimuli, the mycelium reproduces sexually by producing